

Closed-Loop Streaming Synthesis Bounded-Drift, Arbitrary-Length Audio–Video Generation by Treating Chunk Hand-off as Feedback Control

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Abstract

Generating arbitrarily long videos with diffusion transformers requires splitting the temporal dimension into overlapping chunks. Naive open-loop chunk-streaming accumulates exposure-bias drift: each chunk is conditioned on the previously generated latent, compounding distributional errors and causing scene collapse or grain amplification after a few hundred frames. We present **Closed-Loop Streaming Synthesis (CLSS)**, a suite of four post-denoising corrections that close the feedback loop between successive chunks without modifying the transformer weights: calibrated context re-noising, per-channel EMA-tracked AdaIN, frequency-band soft shrinkage, and a dynamic anchor bank. CLSS is implemented as a ComfyUI custom-node pipeline targeting LTX-2.3 22B (a 44 GB audio-video diffusion transformer) on consumer 16 GB VRAM hardware via GGUF 4-bit quantisation and CPU block-streaming. Generation is driven by a reference-parity multi-modal guider (split per-modality classifier-free guidance, modality-isolation guidance, and spatiotemporal guidance via a value-passthrough self-attention skip) that correctly handles ComfyUI’s packed audio-video latent representation, and all streaming noise is seeded for bit-reproducible runs. A two-stage pipeline generates a draft audio-video via CLSS streaming and then refines it using a 3-step distilled-LoRA schedule at $2\times$ spatial resolution.

We report matched text-to-video (T2V) and image-to-video (I2V) experiments: single-scene, ≈ 103 -second clips generated as ten chunks ($\text{new_lf} = 31$, $\text{overlap} = 8$) under an identical seed and the full guidance stack active (video CFG 4.0, audio CFG 7.0, modality 3.0, STG 1.0 on both modalities at block 28, rescale 0.70; four transformer passes per step), differing only in the presence of a guide image. Stage 1 holds video boundary cosine similarity above 0.91 (I2V) and above 0.80 (T2V), with the audio streaming buffer honoured at 1.000 at every seam in both runs and the drift-capped energy anchor holding audio

RMS to a stable floor throughout (0.42–0.50 I2V, 0.59–0.78 T2V). Stage 2 ($2\times$ resolution, 3-step distilled refinement, decoupled audio at denoise fraction 0.4) preserves Stage 1 structure in both runs, with per-window fidelity and video seam similarity in the same ranges. Comparing the two runs resolves a hypothesis the I2V run alone could not test: whether the guide image is responsible for chunk-1 audio coherence and for the phase-lock observed in repetition diagnostics. It is not — T2V achieves comparable chunk-1 audio coherence with no visual anchor and is consistently the *less* phase-locked of the two runs, the opposite of what a stabilising-anchor account predicts. T2V does show one failure mode absent from I2V: audio top-frequency energy grows to $2.1\text{--}2.65\times$ its chunk-1 reference by chunk 7, untouched by any current correction. Total generation time is approximately four hours per ≈ 100 -second clip in either mode on a single 16 GB GPU.

1 Introduction

Modern video diffusion transformers operate on patchified latent representations where temporal resolution is down-scaled $8\times$ relative to pixel frames. Fitting a 30-second clip (81,840 tokens at 22×40 spatial) into a single forward pass requires very large GPUs. The standard remedy is *temporal chunking* via a Streaming Latent Buffer (SLB): generate the video in short overlapping segments, conditioning each on the previous denoised output. Section 2 reviews this mechanism and the exposure-bias problem it introduces.

CLSS closes the feedback loop between chunks by applying four lightweight post-denoising corrections that together bound the closed-loop gain below one, preventing drift from accumulating without modifying the transformer weights. The corrections are applied purely at the boundary between chunks and add negligible computational overhead.

Contributions.

1. Four closed-loop correction modules for temporal chunk streaming: calibrated context re-noising, EMA-tracked per-channel AdaIN, frequency-band soft shrinkage, and a dynamic anchor bank (Sections 3.1–3.4).
2. A linear stability analysis of the feedback loop in latent space, including an empirical per-chunk gain measurement protocol (Section 3.5).
3. A memory-efficient two-stage pipeline for LTX-2.3 22B on 16 GB VRAM using GGUF 4-bit quantisation and CPU block-streaming (Section 5).
4. A reference-parity multi-modal guider for ComfyUI (split per-modality CFG, modality-isolation guidance, and STG via value-passthrough self-attention skip) that correctly handles ComfyUI’s packed audio-video latent representation (Section 5.5).
5. ComfyUI custom nodes implementing the full pipeline with seeded, bit-reproducible streaming noise and per-chunk reproducibility telemetry: unconditional settings dumps, per-frame origin/layout drift event tables, and a phase-lock repetition detector (Sections 5.4, 7.6).

6. Matched ≈ 103 -second T2V and I2V generations, identical seed and guidance stack, streamed as ten chunks each under the active guidance stack, isolating the effect of visual conditioning on audio coherence and phase-lock, and characterising the residual repetition mechanisms at this horizon — including an audio high-frequency growth failure mode present only without a guide image (Sections 6, 7.6).

2 Background

2.1 LTX-2.3 Architecture

LTX-2.3 is a 22-billion-parameter diffusion transformer for joint audio-video synthesis. It operates on a latent space with an $8\times$ temporal downscale (video VAE, causal) and a $160\times 4\times$ audio downscale (audio VAE, 16 kHz, hop length 160, compression factor 4), yielding a video latent rate of $25/8 = 3.125$ lf/s and an audio latent rate of $16000/(160\times 4) = 25.0$ lf/s. The model accepts interleaved video and audio tokens together with Gemma-3 text embeddings and is trained with flow-matching on the rectified-flow schedule.

2.2 Streaming Latent Buffer (SLB)

The SLB provides temporal continuity in chunk streaming. When generating chunk N , the last overlap lf latent frames from chunk $N - 1$ are prepended to the new chunk’s latent, and the denoiser is conditioned on them at noise level τ_c :

$$\tilde{L}_{\text{overlap}} = (1 - \tau_c) \hat{L}_{\text{overlap}} + \tau_c \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, I) \quad (1)$$

Setting $\tau_c = 0$ yields maximal continuity but maximal drift risk; $\tau_c = 1$ decouples the chunks entirely. In LTX’s framework this is implemented as `VideoConditionByLatentIndex` with `strength = 1 - \tau_c`.

2.3 Exposure Bias in Streaming Diffusion

Let $\hat{L}_N$ be the denoised latent of chunk N and $f_\theta(\cdot \mid \hat{L}_{N-1})$ the denoising operator conditioned on the previous chunk. In open-loop streaming:

$$\hat{L}_N = f_\theta(\hat{L}_{N-1} + \varepsilon_N) \quad (2)$$

The transformer was trained on latents drawn from the data distribution, but at inference it receives denoised latents from prior chunks, which lie off-manifold relative to its training distribution. Any distributional shift $\delta_N = \hat{L}_N - L_N^*$ (where L_N^* is the “on-manifold” latent for the same content) is in general amplified by the open-loop gain of the denoising process. Formally, if $g = \|\partial f_\theta / \partial L\|$ evaluated at L_{N-1}^* , then $\|\delta_{N+1}\| \approx g \|\delta_N\|$. When $g > 1$ this gives $\|\delta_N\| \sim g^N \rightarrow \infty$.

In practice g is not a global constant: it depends on the noise level schedule, step count, guidance strength, and the spectral content of the latent. Section 3.5 describes how g is estimated empirically per chunk via a perturbation measurement.

3 Closed-Loop Streaming Synthesis

CLSS adds four closed-loop correction modules that together bound the error accumulation by ensuring $\rho_{\text{closed}} = g \cdot (1 - \beta) \cdot (1 - \tau_c) < 1$.

3.1 Calibrated Context Re-Noising

The overlap latent fed as context to chunk N is re-noised to level τ_c via Eq. (1). This forces the denoiser to actively re-project the boundary context onto the data manifold rather than accepting it verbatim, partially correcting accumulated drift in the SLB.

In practice $\tau_c = 0.05$ provides a strong temporal constraint (boundary cosine similarity > 0.97) while allowing enough distributional repair to prevent drift accumulation. Larger values ($\tau_c = 0.20$, as suggested by some prior work) loosen the constraint too much for 22B GGUF streaming, causing visible motion discontinuities.

3.2 EMA-Tracked Per-Channel AdaIN

A slow exponential moving average (EMA) with rate $\lambda \approx 0.10$ tracks per-channel mean μ_c and standard deviation σ_c across chunks:

$$\mu_c^{(N)} = (1 - \lambda) \mu_c^{(N-1)} + \lambda \hat{\mu}_c^{(N)} \quad (3)$$

$$\sigma_c^{(N)} = (1 - \lambda) \sigma_c^{(N-1)} + \lambda \hat{\sigma}_c^{(N)} \quad (4)$$

where $\hat{\mu}_c^{(N)}, \hat{\sigma}_c^{(N)}$ are the per-channel statistics of chunk N 's new frames. After generation, each chunk's new frames are blended toward the EMA reference via AdaIN with blend factor β :

$$\tilde{L}_N = (1 - \beta) L_N + \beta \text{AdaIN}(L_N \rightarrow \mu^{(N-1)}, \sigma^{(N-1)}) \quad (5)$$

This suppresses fast per-chunk statistical drift while allowing slow intentional scene evolution (lighting changes, colour grading) to pass through.

Two safety caps prevent over-correction:

- **Sigma max drift:** the EMA σ_c is clamped to at most $(1 + \Delta) \times \sigma_c^{(0)}$ (default $\Delta = 0.05$), preventing AdaIN from amplifying late-chunk residual noise.
- **AdaIN max amplification:** per-channel upward scaling is capped at $1.2\times$ the current chunk's std (default), preventing AdaIN from boosting noise in channels that have become quiet.

3.3 Frequency-Band Soft Shrinkage

The latent is decomposed into $B = 4$ frequency bands via 3-D FFT along the (frame, height, width) axes. Each band b receives a continuous gain:

$$g_b^{(c)} = \min \left(1, \left(\frac{E_{\text{ref},b}^{(c)}}{E_b^{(c)}} \right)^{\gamma_b} \right) \in (0, 1] \quad (6)$$

where $E_b^{(c)}$ is the mean squared amplitude of band b in channel c , $E_{\text{ref},b}^{(c)}$ is the corresponding EMA reference, and γ_b is the per-band shrinkage exponent.

Only bands with excess energy relative to their EMA are attenuated. Bands with $\gamma_b = 0$ (the DC band) are never attenuated. Default exponents: $\gamma = (0.0, 0.05, 0.2, 0.3)$, which are conservative enough to preserve scene structure while suppressing temporal flicker.

The EMA is updated with the *corrected* (post-gain) energies, not the raw ones, so the reference tracks intentional scene content rather than accumulated error.

3.4 Dynamic Anchor Bank

A small bank of $M \leq 8$ keyframe anchors provides long-range visual identity conditioning. The bank is seeded from the first frame of chunk 1 and updated when two consecutive chunks trigger a scene-change signal (cosine similarity below threshold $\rho_a = 0.78$ for $P = 2$ consecutive chunks). Additionally, a *forced insertion* every $F_a = 5$ chunks prevents bank stagnation on visually stable sequences where the similarity threshold is never triggered.

Anchors are retrieved via cosine similarity and injected as `VideoConditionByKeyframeIndex` at frame index 0 of the current chunk. Anchors too similar to the current overlap ($\text{sim} > 0.85$) are skipped to avoid duplicating the SLB conditioning. LRU eviction discards the least recently used anchor when the bank is full.

3.5 Stability Analysis

The AdaIN blend (Section 3.2) and context re-noising (Section 3.1) together reduce the magnitude of any per-chunk latent perturbation by a factor $(1 - \beta)(1 - \tau_c)$ before it reaches the next chunk’s denoiser. This gives the *loop gain*:

$$\rho_{\text{loop}} = (1 - \beta) \cdot (1 - \tau_c) \quad (7)$$

The combined closed-loop gain, accounting for the model’s open-loop amplification g , is:

$$\rho_{\text{closed}} = g \cdot \rho_{\text{loop}} \quad (8)$$

Bounded drift requires $\rho_{\text{closed}} < 1$. With $\beta = 0.40$ and $\tau_c = 0.05$, $\rho_{\text{loop}} = 0.57$, providing $\approx 43\%$ gain reduction per chunk before accounting for g .

Note that Eq. (7) is a first-order approximation that treats the two corrections as independent multiplicative attenuators. In practice g is not constant: it depends on the step count, guidance strength, noise schedule, and the spectral properties of the current latent. The analysis therefore provides a stability condition rather than a precise quantitative bound. The frequency-band shrinkage (Section 3.3) provides additional spectral attenuation beyond ρ_{loop} , especially in high-frequency bands, but is not included in the linear gain model because its contribution is content- and chunk-dependent.

Hyperparameter Choices. $\beta = 0.40$ is the largest value that preserves scene-change responsiveness in practice: at $\beta \geq 0.5$ slow colour shifts accumulate due to excessive EMA anchoring. $\tau_c = 0.05$ is a conservative re-noising level that maintains high boundary similarity (> 0.97) while partially correcting SLB distribution shift; larger values ($\tau_c \geq 0.20$) cause visible motion discontinuities on 22B streaming. Together they yield $\rho_{\text{loop}} = 0.57$, which empirically provides adequate margin given that g appears to be below ≈ 1.07 for the tested configuration (Section 7). $\lambda = 0.10$ (EMA rate) was chosen to respond to gradual scene evolution without over-weighting individual chunks; $\gamma = (0.0, 0.05, 0.2, 0.3)$ are conservative exponents that attenuate only bands with clear excess energy.

Per-Chunk Gain Measurement. g is measured empirically by running a second denoising pass with a slightly perturbed overlap latent $\tilde{L}_{\text{overlap}} + \delta$ ($\|\delta\|_F = \epsilon \|\tilde{L}_{\text{overlap}}\|_F$, $\epsilon = 0.01$) and comparing the output perturbation to the input:

$$g_{\text{SLB}} = \frac{\|\Delta_{\text{output}}^{\text{SLB}}\|_F}{\|\delta\|_F} \quad (9)$$

where $\Delta_{\text{output}}^{\text{SLB}}$ is restricted to the last overlap_lf new frames—the portion that becomes the next chunk’s SLB. This measures sensitivity of the boundary output to boundary input perturbations, which is the relevant quantity for SLB-mediated drift propagation. It does not capture drift arising from other channels (e.g., prompt conditioning or attention to non-overlap frames), so g_{SLB} is a lower bound on the true open-loop gain.

4 Algorithm

Algorithm 1 summarises the per-chunk CLSS procedure.

Algorithm 1 CLSS per-chunk generation

Require: Config: $\tau_c, \beta, \lambda, \gamma$; state: SLB, EMA, anchor bank

- 1: $L_{\text{ov}} \leftarrow \text{SLB.read}()$ $\triangleright$ may be empty for chunk 0
 - 2: $\tilde{L}_{\text{ov}} \leftarrow (1 - \tau_c) L_{\text{ov}} + \tau_c \varepsilon$ $\triangleright$ Eq. (1); Sect. 3.1
 - 3: $A \leftarrow \text{anchor_bank.retrieve}(\text{top-}m \text{ by cosine sim})$ $\triangleright$ Sect. 3.4
 - 4: $L_N \leftarrow \text{Denoise}(\tilde{L}_{\text{ov}}, A, \text{prompt}_N)$
 - 5: $L_N^{\text{new}} \leftarrow L_N[\text{overlap_lf} :]$ $\triangleright$ drop overlap region
 - 6: $L_N^{\text{new}} \leftarrow \text{AdaIN_blend}(L_N^{\text{new}}, \text{EMA}, \beta)$ $\triangleright$ Sect. 3.2
 - 7: $L_N^{\text{new}} \leftarrow \text{band_soft_shrink}(L_N^{\text{new}}, \text{band_EMA}, \gamma)$ $\triangleright$ Sect. 3.3
 - 8: $\text{EMA.update}(L_N^{\text{new}}, \lambda)$; $\text{band_EMA.update}(L_N^{\text{new}}, \lambda)$
 - 9: $\text{anchor_bank.update}(\text{last frame of } L_N^{\text{new}})$ $\triangleright$ Sect. 3.4
 - 10: $\text{SLB.push}(L_N^{\text{new}}[-\text{overlap_lf} :])$
 - 11: **return** L_N^{new}
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5 Implementation

5.1 Memory Budget for 16 GB VRAM

LTX-2.3 22B in BF16 occupies ≈ 44 GB. Three complementary strategies fit it within 16 GB VRAM:

1. **GGUF 4-bit quantisation** (Q4_K_S): reduces the on-disk size to ≈ 11 GB. De-quantization to BF16 at load time requires ≈ 44 GB of pinned CPU RAM.
2. **CPU block-streaming**: all transformer blocks reside in pinned CPU RAM; only 2 blocks are resident on GPU at a time (≈ 5 GB model weights on GPU), with the next block pre-fetched asynchronously via a dedicated CUDA stream.
3. **Temporal chunking (CLSS)**: latent memory is $O(\text{overlap_lf})$ instead of $O(\text{total_frames})$. Processed chunk latents are immediately offloaded to CPU after each chunk.

Combined peak VRAM budget at 512×768 resolution:

Component	Peak VRAM
Model blocks ($2 \times$ BF16, streaming)	≈ 5 GB
Activations + latents (per chunk)	≈ 4 GB
VAE / text encoder (loaded/freed)	≈ 2 GB
Total peak	≈ 11 GB

5.2 Two-Stage Pipeline

Stage 1 (Open-resolution streaming). CLSS streaming generates the full video at a coarse latent resolution ($H_L \times W_L$ latent grid, typically 11×20 for 704×512 output). Each chunk runs the full 20–30-step LTX scheduler. CLSS corrections are applied between chunks as per Algorithm 1.

Stage 2 (Distilled-LoRA refinement). Stage 1 operates at 704×512 output resolution and 20 denoising steps, which is sufficient for temporal structure and scene content but produces visible quantisation artefacts and reduced texture detail, particularly after GGUF 4-bit dequantisation at the low-frequency latent scale. Stage 2 addresses this by upsampling and refining the output.

The Stage 1 output is spatially upsampled $2\times$ by `LTXVLatentUpsampler`, yielding a 22×40 latent grid corresponding to 1408×1024 pixel resolution. Stage 2 then refines the full video in temporal chunks using a 3-step distilled-LoRA schedule with initial noise level $\sigma_0 \approx 0.909$. Each chunk’s new frames are seeded from the upscaled Stage 1 latent with σ_0 -scaled noise:

$$x_{\text{start}} = \sigma_0 \varepsilon + (1 - \sigma_0) L_N^{\text{S1}} \quad (10)$$

This formulation anchors Stage 2 to the coarse temporal structure produced by Stage 1—preserving scene layout, motion trajectories, and chunk boundaries established by

CLSS—while allowing the 3-step refinement to add local high-frequency detail. Stage 1 alone could in principle be generated at $2\times$ resolution, but this would quadruple token count and exceed VRAM budget under CPU block-streaming.

Key differences from Stage 1:

- AdaIN and spectral-shrinkage corrections are *disabled*: Stage 2 refinement is structurally anchored to the S1 latent, so distributional drift correction is unnecessary.
- A global pre-generated noise field is sliced per chunk (`_SlicedNoise`) to prevent grain seams at chunk boundaries—a single coherent noise tensor ensures that the per-chunk noise samples tile seamlessly when concatenated.
- Audio is refined in parallel with the SLB at τ_c using the Stage 2 audio tail as context.

5.3 Audio Streaming

Audio and video share the same temporal chunk boundaries. Audio length accounting is anchored to *pixel time*: the causal video VAE maps a chunk of `new_lf` latent frames to $(\text{new_lf} - 1) \cdot 8 + 1$ pixel frames for the first chunk of a sequence and $\text{new_lf} \cdot 8$ pixel frames for every continuation chunk. The audio rate is taken from the chunk template, $r = \text{new_af} / ((\text{new_lf} - 1) \cdot 8 + 1)$ audio frames per pixel frame, and the SLB length follows the same rate:

$$\text{overlap_af} = \text{round}(\text{overlap_lf} \cdot 8 \cdot r) \quad (11)$$

With `overlap_lf` = 8 and the 13-lf/102-af template this gives `overlap_af` = 67. (An earlier per-chunk accounting ignored the causal first-frame discount, undercounting every continuation chunk by $\approx 0.28\text{s}$ and desynchronising audio from video by $\approx 2\text{s}$ over seven chunks; the pixel-time ledger above eliminates the drift exactly.) The effective video overlap is additionally clamped to `new_lf`: the SLB can never carry more frames than a chunk produces, and unclamped values were measured to silently discard motion at every seam.

Drift-capped energy anchoring. Un-anchored streaming audio is a divergent loop in both directions: runs were measured with RMS growing +67% and low-mid spectral bands tripling over ten chunks. A per-chunk anchor corrects the new audio’s global RMS and per-channel DC toward slowly-moving EMA targets whose drift from the chunk-1 origin is hard-capped at $\pm\Delta_\sigma$ (the same $\pm 5\%$ used for the video EMA), before the chunk’s tail is fed forward as the next SLB/reference. Anchoring *before* feed-forward is what makes the audio SLB safe at any sequence length. A fixed-forever target was tried first and caused repetition (the model re-rendered a stabilised loop); the capped EMA restores bounded evolution. On multi-scene runs the anchor is re-baselined at scene changes; this is measurably a ratchet (each re-baseline legitimises accumulated loudness creep: $0.599 \rightarrow 0.699 \rightarrow 0.823$ over three scenes in one run) and remains a known limitation.

Reference-audio conditioning. Each continuation chunk also receives a reference-audio window (`ref_audio` tokens at negative temporal RoPE) taken from a rolling tail of the accumulated output. Its length is decoupled from the video overlap (default 67af $\approx$ 2s; an implicit coupling to `overlap_af` starved it at small overlaps and measurably collapsed within-chunk audio coherence from 0.91 to 0.27). A small noise fraction is blended into the reference, ramping 0.05/chunk and capped at 0.10 (an earlier hard-coded 0.35 cap injected audible hiss).

Context envelope flattening. The fed-forward audio context (SLB tail and reference window) is optionally flattened in loudness: each frame is rescaled toward the window-mean RMS with per-frame gain clamped to $[0.6, 1.25]$. Motivation is a measured metronome: the audio energy peak locked to the same frame (102/109) for eight consecutive chunks — the crescendo in each fed-forward tail re-seeded the same loudness arc every chunk. Flattening the *context only* (output audio is untouched) broke the phase lock in A/B runs. The gain cap is asymmetric by design: an unbounded first version amplified the noise floor of quiet frames ($\times 2.98$ observed) and injected persistent hiss.

Boundary smoothing. At each audio SLB boundary, a linear blend over ± 2 audio latent frames is applied in latent space before VAE decoding. This eliminates click artifacts that arise from the abrupt splice between the overlap tail and the new chunk, which would otherwise be visible as discontinuities in the spectrogram.

5.4 ComfyUI Node Integration

Five custom nodes implement the full pipeline inside ComfyUI:

1. **CLSSConfig:** exposes the CLSS hyperparameters worth exposing (τ_c, β , overlap); validated internals are fixed.
2. **CLSSScenePrompts:** multi-scene text input via `---` delimiters; Gemma-enhanced per scene. Output is a flat CONDITIONING list with one entry per scene, proportionally assigned to chunks. With one prompt per chunk this doubles as a per-chunk *object-identity* anchor (see Section 8).
3. **CLSSStreamingSampler:** Stage 1 streaming sampler. Accepts any ComfyUI GUIDER/SAMPLER/SIGMAS/NOISE/LATENT combination. Emits an unconditional settings dump, per-chunk telemetry, end-of-run trend summaries with per-chunk localisation, per-frame origin/layout drift event tables, and a phase-lock repetition detector.
4. **CLSSStage2:** Stage 2 refinement sampler with consistent per-chunk noise slicing and a decoupled audio-refinement pass.
5. **CLSSAVGuiderV2:** reference-parity multi-modal guider (Section 5.5). Its predecessor (**CLSSAVGuider**, a `sampler_cfg_function` patch) is retained but inert on current ComfyUI and warns accordingly.

All stochastic inputs are seeded: Stage-1 video *and* audio noise are pre-generated as full-sequence fields sliced per chunk (distinct yet coherent noise per chunk, bit-reproducible runs), and every fallback noise path uses a generator derived deterministically from the seed and chunk position.

5.5 Multi-Modal Guidance and the Packed-Latent Pitfall

The reference pipeline guides each modality independently (`MultiModalGuider`):

$$\text{pred} = c + (s_{\text{cfg}} - 1)(c - u) + s_{\text{stg}}(c - p) + (s_{\text{mod}} - 1)(c - m) \quad (12)$$

per modality, followed by a rescale $\text{pred} *= r \sigma_c / \sigma_{\text{pred}} + (1 - r)$, where u is the text-unconditional prediction, m the modality-isolated prediction (both audio $\leftrightarrow$ video cross-attentions skipped), and p the STG-perturbed prediction. The STG perturbation skips *self*-attention in designated blocks (block 28 for LTX-2.3) by replacing the attention output with the raw value projection ($\text{out} = W_V x$, gating and output projection unchanged) — a value-passthrough, not a zero-out. Defaults follow the reference: video cfg 4–4.5, audio cfg 7, modality 3, STG 1 for both modalities, rescale 0.7; four transformer passes per step.

The packed-latent representation. ComfyUI’s `CFGGuider.sample()` packs nested audio-video latents into a *single flat* $[B, 1, N_v + N_a]$ tensor before sampling and unpacks only after. A guider that tries to detect the audio-video case with `isinstance(x, NestedTensor)` inside `predict_noise` therefore never sees a nested tensor and falls back to plain shared CFG — and it does so silently, producing no error and no log line. The guider must instead capture the per-modality shapes in `sample()` *before* packing, detect the packed $[B, 1, N]$ representation in `predict_noise`, unpack each pass output, apply Eq. (12) per modality, and repack. Two diagnostics guard against a silent regression: the guider announces the active branch and its pass count once per chunk (`passes/step=4` for the four-pass configuration), and it logs the STG and modality difference norms so a zero-valued (inert) perturbation is visible immediately. A useful invariant follows from the pass count: because each guidance term costs a transformer pass, a guidance configuration that does not change per-step wall-clock time is not actually running — on 13-lf chunks the four-pass stack costs $\approx 17.3\text{s/it}$ against $\approx 8.0\text{s/it}$ for plain CFG. Two engineering rules generalise: guidance paths must announce which branch is active, loudly and unconditionally; and a guidance change that does not change step time did not happen.

6 Experiments

Two generation experiments share the same CLSS configuration, seed, and active guidance stack but differ in the first-frame conditioning: a text-to-video (T2V) run with no image prior (Section 6.2), and an image-to-video (I2V) run conditioned on a single guide frame ($1 \times 11 \times 20$ latent, Section 6.3). Both stream ten Stage-1 chunks

(new_lf = 31, overlap = 8) to 310 latent frames (≈ 103 s) and refine through the same seven-window Stage 2 pass.

6.1 Shared Configuration

Table 1: Shared configuration. Guidance values are those actually active during sampling (verified by the guider’s per-chunk **passes/step** and difference-norm telemetry, Section 5.5). The Stage-2 geometry is selected automatically from the VRAM token budget.

Parameter	Value
Model	LTX-2.3 22B (Q4_K_S GGUF)
Resolution	704×512 (Stage 1), 1408×1024 (Stage 2)
τ_c	0.05
β	0.40
λ	0.10
Δ_σ	0.05
F_a	3 (auto, for 10 chunks)
γ	(0.0, 0.05, 0.2, 0.3)
Video guidance	CFG 4.0, STG 1.0, modality 3.0
Audio guidance	CFG 7.0, STG 1.0, modality 3.0
STG block / rescale	28 / 0.70 (four passes/step)
Stage 1 steps	20
Stage 2 steps	3 (distilled), denoise (audio) 0.4
VRAM (GPU)	15.61 GB (RTX-class)
T2V first-frame	Text only
I2V first-frame	Guide image, latent [1, 128, 1, 11, 20]
Stage 1 (both)	10 chunks, new_lf = 31, overlap = 8, 1 scene
Stage 2 (both)	7 windows, fpc = 45, overlap = 15 (decoupled audio)

6.2 Experiment 1: Text-to-Video (T2V)

No image is provided; Stage 1 chunk 1 begins from pure noise conditioned on the text prompt alone. The clip is a single prompted scene streamed as ten chunks (new_lf = 31, overlap = 8), assembling to 310 latent frames (≈ 103 s), under the identical seed and guidance stack as the I2V run (Table 1).

6.2.1 Guidance Activity

As in I2V, every chunk shows **passes/step**=4 with non-zero STG and modality difference norms (Table 2). Chunk 1’s audio modality term is again the outlier ($m_{\text{aud}} = 75.8$ versus 5.9–9.6 thereafter) — the same mechanism as I2V (no audio SLB or reference window on the first chunk), now confirmed independent of the guide image.

Table 2: T2V Stage 1 guidance difference norms (step 1 of each chunk).

Chunk	STG (vid)	STG (aud)	Mod (vid)	Mod (aud)
1	160.1	32.7	26.3	75.8
2	127.7	8.9	44.6	7.5
3	168.7	8.1	46.6	7.7
4	141.1	6.7	40.9	9.6
5	212.1	9.0	45.8	6.0
6	185.3	10.0	47.2	5.9
7	134.8	9.4	50.0	5.9
8	147.4	7.8	43.1	8.5
9	203.8	11.0	49.2	6.5
10	270.9	8.6	56.7	5.8

6.2.2 Stage 1 Results

Table 3: T2V Stage 1 per-chunk video metrics (10 chunks).

Chunk	Boundary sim $\uparrow$	Intra sim $\uparrow$	Identity sim	ρ_{loop}
1/10	N/A	0.4043	1.0000 (ref)	0.570
2/10	0.7974	0.1875	0.7974	0.570
3/10	0.8993	0.3824	0.3824	0.570
4/10	0.8226	0.2927	0.8226	0.570
5/10	0.9559	0.2950	0.9559	0.570
6/10	0.9324	0.2156	0.7823	0.570
7/10	0.8330	0.1957	0.4485	0.570
8/10	0.8725	0.3554	0.8725	0.570
9/10	0.8422	0.3186	0.7492	0.570
10/10	0.9677	0.6978	0.7403	0.570

Without a guide image, video boundary continuity is weaker and more variable than I2V: it ranges 0.797–0.968 (versus I2V’s 0.910–0.967), with its floor again at chunk 2 — the same first SLB-conditioned hand-off — but 11 points lower than I2V’s floor at the identical seam. Origin similarity to the scene reference decays much further over the run (0.216 $\rightarrow$ 0.108, range 0.141, versus I2V’s 0.259 $\rightarrow$ 0.242, range 0.106): with nothing pinning the opening frame, content wanders farther from where it started. Identity similarity to the anchor bank also dips lower, to 0.382 at chunk 3 (versus I2V’s floor of 0.747). Unlike I2V, where the largest structural events all clustered at the chunk 1 $\rightarrow$ 2 seam, T2V’s largest coarse-layout drops are spread across the whole run — at chunks 1, 3, 6, 7, 9, and 10, with the single deepest ($\Delta = -0.382$) landing near the very end (chunk 10, $t \approx 98$ s) rather than at the first seam. Origin-similarity drops do still concentrate at chunk 2 (five of the six largest events), and that single

deepest origin event ($\Delta = -0.189$) is if anything slightly larger than I2V’s equivalent ($\Delta = -0.162$) at the same seam.

Table 4: T2V Stage 1 AdaIN corrections and band gains.

Chunk	$\Delta\mu$	$\Delta\sigma$	g_0	g_1	g_2	g_3
1	+0.0000	+0.0000	1.0000	1.0000	1.0000	1.0000
2	+0.0084	−0.0051	1.0000	0.9995	0.9984	0.9966
3	+0.0027	−0.0445	1.0000	0.9939	0.9725	0.9573
4	+0.0034	−0.0330	1.0000	0.9971	0.9951	0.9769
5	−0.0022	−0.0634	1.0000	0.9922	0.9679	0.9468
6	+0.0128	−0.0405	1.0000	0.9972	0.9858	0.9697
7	−0.0050	−0.0350	1.0000	0.9958	0.9909	0.9786
8	+0.0058	−0.0222	1.0000	0.9980	0.9924	0.9815
9	+0.0037	−0.0334	1.0000	0.9977	0.9829	0.9539
10	+0.0015	−0.0298	1.0000	0.9988	0.9956	0.9857

Every T2V chunk receives a *downward* std correction ($\Delta\sigma < 0$ throughout), in contrast to I2V’s mixed-sign pattern (Table 9): without an image-conditioned first chunk to compress toward, each text-only chunk’s raw std runs slightly hot relative to the EMA reference and AdaIN consistently pulls it back down. Spectral shrinkage is correspondingly more active than in I2V, with band-3 gains regularly in the 0.95–0.96 range (vs. I2V’s mostly ≥ 0.98) and a low of 0.9468 at chunk 5.

The anchor bank grows identically to I2V, from 1 to 5 entries at the same frame positions (`bank_frame_ids` = [30, 92, 123, 216, 309]): because the bank’s forced-insertion cadence ($F_a = 3$) is driven by chunk position, not content, the two runs — sharing chunk boundaries and seed — produce the same growth schedule despite entirely different video content.

6.2.3 Stage 1 Audio

Table 5: T2V Stage 1 per-chunk audio metrics.

Chunk	WC start	WC end	env_corr	RMS	Bnd sim	SLB
1	0.948	0.969	—	0.775	N/A	N/A
2	0.977	0.952	−0.273	0.723	0.9496	1.000
3	0.927	0.915	0.040	0.605	0.9658	1.000
4	0.883	0.901	0.350	0.565	0.9771	1.000
5	0.942	0.925	0.564	0.630	0.9658	1.000
6	0.934	0.907	0.913	0.613	0.9699	1.000
7	0.875	0.854	−0.486	0.600	0.9638	1.000
8	0.951	0.948	−0.342	0.613	0.9782	1.000
9	0.942	0.927	0.789	0.611	0.9635	1.000
10	0.789	0.904	0.847	0.591	0.9731	1.000

Chunk 1 audio, with no guide image and no audio buffer, is already coherent (within-chunk $0.948 \rightarrow 0.969$) — closely matching I2V’s chunk 1 coherence ($0.805 \rightarrow 0.985$) rather than collapsing. This resolves the comparison deferred in Section 6.3.3: the guide image is *not* the source of chunk-1 audio coherence, since T2V achieves comparable coherence with no visual anchor at all. The seeded Stage-1 audio noise field, shared by both runs, is the more likely explanation; chunk-1 audio coherence here reflects the audio generation process itself rather than cross-modal grounding. Audio boundary similarity is high throughout (0.950–0.978), even better than I2V’s 0.910–0.950. The absolute loudness level is also substantially higher than I2V’s (0.59–0.78 RMS versus 0.42–0.50), and the drift-capped anchor again holds it: the EMA target settles at $\text{ema_rms} \approx 0.743$ from chunk 3 onward and every raw chunk RMS stays close to that floor.

The loudness-envelope correlation is qualitatively different from I2V. Where I2V’s `env_corr` climbs smoothly into a 0.77–0.95 lock from chunk 3 onward, T2V’s swings between strongly negative (−0.49, −0.34, −0.27) and strongly positive (0.91, 0.85, 0.79) values with no monotonic settling — the phase-lock detector confirms this is looser, not tighter: the audio energy peak covers only 3/10 chunks within ± 1 of its mode (versus I2V’s 5/10), and the coarse video-layout argmin covers only 2/10 (versus I2V’s 4/10). Taken together with the chunk-1 audio result above, the guide image appears to correlate with *more* metronomic repetition, not less — the opposite of a naive “visual anchor stabilises everything” hypothesis. This is examined further in Section 7.6.

6.2.4 Stage 2 Results

Table 6: T2V Stage 2 per-window metrics (7 windows).

Window	Bnd sim $\uparrow$	Intra sim	Fid first	Fid last	Aud_bnd
1/7	N/A	-0.2311	0.9014	0.9269	N/A
2/7	0.8362	0.5186	0.9636	0.9469	0.7539
3/7	0.9560	0.2844	0.9490	0.9572	0.8711
4/7	0.9401	0.4147	0.9334	0.9460	0.8003
5/7	0.9523	0.4510	0.9478	0.9580	0.8051
6/7	0.8668	0.6114	0.9247	0.9329	0.8456
7/7	0.8288	0.2344	0.9497	0.7800	0.8440

Window 1’s intra similarity is *negative* (-0.231), sharply unlike I2V’s positive 0.610 for the same window: the unanchored T2V chunk 1 (Table 3, intra 0.404) carries into Stage 2 as a first window whose start and end frames are anti-correlated, whereas I2V’s guide image keeps its first window internally coherent. From window 2 onward video seam continuity behaves similarly to I2V (0.83–0.96, weakest at windows 2 and 7). Fidelity to the Stage-1 latent shows the same end-of-run erosion as I2V, bottoming at an almost identical value in the final window (0.780 versus I2V’s 0.788) — the softening in the last ≈ 15 s appears to be a Stage-2 property largely independent of the conditioning mode. The decoupled audio pass, however, seams noticeably *better* than I2V here (0.754–0.844 versus I2V’s 0.594–0.721), worst again at the first seam (15 s) — consistent with the higher raw audio boundary similarity already seen in Stage 1 (Table 5).

6.3 Experiment 2: Image-to-Video (I2V)

A single guide image is encoded to a latent of shape $[1, 128, 1, 11, 20]$ and provided as first-frame conditioning to Stage 1 chunk 1. The clip is a single prompted scene streamed as ten chunks ($\text{new_lf} = 31$, $\text{overlap} = 8$), assembling to 310 latent frames (≈ 103 s). All CLSS hyperparameters and the guidance stack are those of Table 1.

6.3.1 Guidance Activity

The guider emits `passes/step=4` and its STG and modality difference norms once per chunk (Table 7); both stay well away from zero on every chunk, so no step fell back to plain shared CFG. Chunk 1 is an outlier on the audio modality term ($m_{\text{aud}} = 72.3$ versus 5.2–6.4 thereafter): chunk 1 has neither an audio SLB nor a reference-audio window, so the modality-isolated audio prediction has nothing to reconcile against and diverges further from the joint prediction; once conditioned, every later chunk collapses to a tight band.

Table 7: I2V Stage 1 guidance difference norms (step 1 of each chunk). STG measures the self-attention-skip perturbation; modality measures the cross-attention-skip perturbation. All non-zero $\Rightarrow$ guidance active.

Chunk	STG (vid)	STG (aud)	Mod (vid)	Mod (aud)
1	317.9	26.7	47.9	72.3
2	126.0	8.5	36.9	5.4
3	167.3	10.7	39.0	6.2
4	186.1	8.6	42.0	5.6
5	204.1	8.3	41.2	5.2
6	218.0	8.6	55.8	5.5
7	241.5	10.1	44.8	5.6
8	261.4	10.4	65.2	5.5
9	231.8	10.7	47.2	6.4
10	190.5	9.4	69.6	6.1

6.3.2 Stage 1 Results

Table 8: I2V Stage 1 per-chunk video metrics (10 chunks). Boundary_sim compares the last overlap frame of the previous chunk against the first new frame; identity_sim compares the chunk against its nearest anchor-bank entry.

Chunk	Boundary sim $\uparrow$	Intra sim $\uparrow$	Identity sim	ρ_{loop}
1/10	N/A	0.6829	1.0000 (ref)	0.570
2/10	0.9097	0.6562	0.9097	0.570
3/10	0.9546	0.8310	0.8310	0.570
4/10	0.9513	0.8094	0.9513	0.570
5/10	0.9613	0.6755	0.9613	0.570
6/10	0.9599	0.5393	0.7466	0.570
7/10	0.9477	0.6978	0.8103	0.570
8/10	0.9300	0.7294	0.9300	0.570
9/10	0.9657	0.9020	0.8748	0.570
10/10	0.9669	0.8352	0.8427	0.570

Video boundary continuity holds between 0.910 and 0.967 across all nine chunk transitions, with the floor at chunk 2 — the first SLB-conditioned hand-off, between the unconditioned chunk 1 and its first continuation. The largest structural events of the whole run cluster at exactly this seam ($\approx 10.3\text{--}10.7\text{ s}$): the run’s single deepest coarse-layout drop ($\Delta = -0.332$) and three of its four largest per-frame origin-similarity drops all land at the chunk 1 $\rightarrow$ 2 boundary. Global origin similarity to the scene reference is nonetheless contained over the full run ($0.259 \rightarrow 0.242$, range 0.106), in contrast to the monotonic walk-away drift seen at wider overlaps in earlier open-loop testing. Identity similarity to the anchor bank erodes gradually ($0.910 \rightarrow 0.843$) with

a local minimum of 0.747 at chunk 6, indicating some content departure from the banked keyframes without runaway divergence.

Table 9: I2V Stage 1 AdaIN corrections and band gains (Band 0 is the DC band, $\gamma_0 = 0$, $\text{gain} \equiv 1$). $\Delta\mu$, $\Delta\sigma$ are the applied per-chunk AdaIN mean/std deltas.

Chunk	$\Delta\mu$	$\Delta\sigma$	g_0	g_1	g_2	g_3
1	+0.0000	+0.0000	1.0000	1.0000	1.0000	1.0000
2	+0.0021	−0.0017	1.0000	0.9963	0.9905	0.9906
3	+0.0064	−0.0092	1.0000	0.9971	0.9926	0.9899
4	+0.0082	−0.0173	1.0000	0.9965	0.9883	0.9890
5	+0.0049	−0.0059	1.0000	0.9965	0.9907	0.9930
6	+0.0075	−0.0161	1.0000	0.9969	0.9862	0.9886
7	+0.0096	+0.0020	1.0000	0.9976	0.9897	0.9938
8	−0.0053	−0.0853	1.0000	0.9863	0.9686	0.9615
9	+0.0022	−0.0313	1.0000	0.9949	0.9816	0.9827
10	+0.0020	−0.0365	1.0000	0.9934	0.9780	0.9807

Most chunks receive small AdaIN corrections and mild spectral shrinkage. The exception is chunk 8, whose raw pre-correction std reached 1.125 (vs ≈ 0.95 elsewhere): AdaIN pulls it back to 0.955 ($\Delta\sigma = -0.085$) and the spectral shrinkage bites hardest here (band-3 gain 0.9615), the two corrections jointly absorbing a localised energy spike before it can propagate through the SLB. This is the correction stack operating as designed — a single anomalous chunk is renormalised rather than fed forward.

The anchor bank grows from 1 to 5 entries over the run (`bank.frame_ids = [30, 92, 123, 216, 309]`), driven by the automatic forced-insertion cadence ($F_a = 3$ for a 10-chunk run) since the single-scene content rarely trips the similarity threshold. Because the ComfyUI sampler currently uses the bank for identity telemetry only (the retrieval-injection conditioning path of Section 3.4 is present in the reference library but not yet wired into this sampler), the bank here measures identity drift rather than correcting it.

6.3.3 Stage 1 Audio

Table 10: I2V Stage 1 per-chunk audio metrics. Within-chunk similarity is reported at the start and end of each chunk; `env_corr` is the loudness-envelope correlation with the previous chunk (> 0.7 flags a repeating gesture); SLB is the audio streaming-buffer honoured fraction.

Chunk	WC start	WC end	env_corr	RMS	Bnd sim	SLB
1	0.805	0.985	—	0.502	N/A	N/A
2	0.957	0.931	0.145	0.466	0.9096	1.000
3	0.990	0.955	0.768	0.438	0.9415	1.000
4	0.986	0.923	0.933	0.445	0.9383	1.000
5	0.978	0.869	0.892	0.432	0.9447	1.000
6	0.954	0.811	0.909	0.429	0.9449	1.000
7	0.941	0.800	0.904	0.427	0.9496	1.000
8	0.934	0.795	0.945	0.424	0.9427	1.000
9	0.930	0.785	0.941	0.421	0.9486	1.000
10	0.932	0.790	0.923	0.423	0.9417	1.000

The audio streaming buffer is honoured exactly (1.000) at every seam, and the audio boundary similarity is stable at 0.91–0.95 throughout. The drift-capped EMA energy anchor holds the per-chunk RMS within 0.42–0.50 and settles toward its anchored target ($\text{ema_rms} \approx 0.463$) rather than climbing: the divergent loudness growth measured on un-anchored audio does not appear. Under the guide image, chunk 1 audio is already coherent (within-chunk $0.805 \rightarrow 0.985$) despite having no audio buffer. Section 6.2.3 tests whether this is attributable to the visual anchor via cross-modal attention or to the seeded audio noise alone, by repeating the identical comparison with no guide image present.

Two repetition signatures remain. First, the end-of-chunk within-chunk similarity declines monotonically across the run (0.931 at chunk 2 to 0.785 at chunk 9): chunk *starts* stay coherent, but each chunk drifts further from its own opening by the time it ends as the run proceeds. Second, the loudness-envelope correlation with the previous chunk locks into 0.77–0.95 from chunk 3 onward — the same loudness gesture recurs each chunk. Envelope flattening (Section 5.3) reduces this correlation but does not eliminate it; the phase-lock detector places the audio energy peak at relative frame 7 in most chunks (mode 7). Both are characterised further in Section 7.6.

6.3.4 Stage 2 Results

Stage 2 refines the $2\times$ -upsampled latent in seven overlapping windows ($\text{fpc} = 45$, $\text{overlap} = 15$) with the audio pass *decoupled*: the video refines against frozen Stage-1 audio, then the audio is refined separately against the refined video at denoise fraction 0.4.

Table 11: I2V Stage 2 per-window metrics (7 windows). Fid is video fidelity to the Stage-1 latent (first / last frame of the window); Aud_bnd is the audio window-boundary similarity.

Window	Bnd sim $\uparrow$	Intra sim	Fid first	Fid last	Aud_bnd
1/7	N/A	0.6095	0.9771	0.9466	N/A
2/7	0.9183	0.7797	0.9530	0.9421	0.5937
3/7	0.9348	0.7247	0.9433	0.8949	0.6867
4/7	0.8455	0.4013	0.8964	0.9562	0.6512
5/7	0.9543	0.5582	0.9481	0.9379	0.6273
6/7	0.9385	0.8011	0.9300	0.9009	0.7115
7/7	0.8456	0.7035	0.8556	0.7879	0.7207

Video seam continuity ranges 0.85–0.95, weakest at windows 4 and 7 (both 0.846). Window 4 ($t \approx 44.7$ – 59.3 s) is the weakest window of the Stage-2 pass by a clear margin (intra similarity 0.401 against 0.56–0.80 elsewhere). Fidelity to the Stage-1 latent erodes toward the end of the run: the end-of-window fidelity falls from 0.947 in window 1 to 0.788 in the final window ($t \approx 88.7$ – 103.3 s), i.e. the refinement softens away from the Stage-1 structure most in the last ≈ 15 s.

The decoupled audio pass at denoise 0.4 *polishes* rather than rebuilds: the refined audio stays close to its Stage-1 seed at every window (seed similarity 0.79–0.81). Even so, the audio window-boundary similarity is only 0.59–0.72 (worst at the first seam, 15 s) despite the audio SLB being honoured at 1.000 at every window — the SLB pins the seam *frame*, but two independently denoised windows still disagree in loudness and character across it. The seam similarity does trend upward over the run ($0.594 \rightarrow 0.721$), tracking the Stage-1 audio settling into its anchored energy floor; the single-window-vs-multi-window rebuild trade-off behind this number is discussed in Section 7.6. The assembled refined latents are [1, 128, 310, 22, 40] (video, std 0.950) and [1, 8, 2593, 16] (audio, std 0.521), with no NaN/Inf.

6.4 Computational Overhead

Table 12: Wall-clock timing for the I2V run on a 16 GB GPU (RTX-class, CPU block-streaming); the T2V run’s per-chunk/per-window timings match to within a few percent throughout (see text) and are omitted for brevity. Stage 1 runs four transformer passes per step (split CFG, modality, STG); each Stage 2 window runs a video pass and a separate decoupled audio pass.

Stage / pass	Chunk	Steps	s/it	Time	Tokens
S1	1/10	20	38.7	≈ 12.9 min	—
S1	2–10/10	20	47.8–49.2	≈ 16.1 min each	—
S2 video	1/7	3	79.6	≈ 4.0 min	39,600
S2 video	2–7/7	3	118–121	≈ 5.9 min each	$\approx 52,000$
S2 audio	each	3	79.7–121	$\approx$ matches video pass	—

Total generation time: Stage 1 ≈ 158 min + Stage 2 ≈ 79 min $\approx$ **237 min** (≈ 4 h) for a ≈ 103 -second clip. The dominant cost driver is the four-pass guidance in Stage 1 — each denoising step now runs four transformer passes (conditional, unconditional, modality-isolated, STG-perturbed) rather than one — combined with ten chunks at `new_lf` = 31. In Stage 2 the per-window cost more than doubles after window 1 (79.6 s/it $\rightarrow$ 118–121 s/it) despite the video token count rising only $\approx 31\%$: as the run proceeds, an increasing share of the model is pushed to CPU (offloaded weights grow from ≈ 6.6 GB to ≈ 9.1 GB across the pass), so the slowdown is disproportionate to the token increase and is an offload-pressure effect rather than a compute-scaling one.

The T2V run (Stage 1 ≈ 158 min, Stage 2 ≈ 80 min, total ≈ 238 min) costs the same as I2V to within rounding: per-chunk and per-window s/it match to a few percent at every position (e.g. Stage-2 window 2: 121.9 s/it T2V vs. 121.5–122.8 s/it I2V), confirming that the guide image changes neither the token count nor the compute cost — only the conditioning. The Stage-2 offload-pressure slowdown reproduces at the same windows in both runs, supporting that it is a property of the refinement pipeline rather than of run content.

7 Discussion

7.1 Effect of Chunk Length

Both runs used `new_lf` = 31 (≈ 10 s per chunk), exceeding the 21-frame safe limit established in earlier open-loop testing, streamed as ten chunks under the active guidance stack. In neither run does boundary similarity degrade with chunk index: I2V holds 0.910–0.967 with its floor at the first SLB-conditioned hand-off (chunk 2), and T2V holds 0.797–0.968 with its floor at the same seam — the corrected loop does not compound error over this horizon in either conditioning mode, though T2V’s floor

sits noticeably lower. Spectral shrinkage engages more in T2V: band-3 gains regularly fall to 0.95–0.96 (I2V mostly stays ≥ 0.98), consistent with every T2V chunk needing a downward AdaIN std correction (Table 4) where I2V’s corrections are mixed-sign. In both runs the correction stack responds to measured excess rather than applying a fixed schedule — it engages hardest exactly where a chunk’s raw statistics run hottest (I2V chunk 8; T2V nearly every chunk).

7.2 Identity Tracking and Boundary Self-Consistency

Chunk 2’s `identity_sim` (0.9097) equals its `boundary_sim` exactly. This is not a coincidence but a *self-consistency check*: the anchor was seeded from chunk 1’s last frame, and the SLB passes that same frame as the first latent frame of chunk 2’s conditioning, so the anchor query and the boundary query compare the same source frame (chunk 1 frame 30) against the same target frame (chunk 2 frame 0). Both metrics agreeing validates that the anchor bank and the SLB operate on the same reference, and that the τ_c re-noising at the boundary introduces only minimal distributional shift. For later chunks the bank holds multiple entries and the identity query selects a different anchor than the boundary, so the two metrics diverge, as expected (e.g. chunk 6: identity 0.747 vs boundary 0.960). The same self-consistency check holds in the unrelated T2V run at the same chunk (identity 0.7974 = boundary 0.7974, Table 3), confirming the mechanism is a property of the anchor/SLB bookkeeping rather than of any particular content.

7.3 Audio Coherence and the Role of Visual Context

A natural hypothesis is that a guide image substitutes for the absent chunk-1 audio buffer as a coherence anchor: in LTX-2.3’s joint audio-video transformer, video and audio tokens share self-attention layers, so a strong single-frame visual prior could constrain audio generation implicitly through cross-modal attention. The I2V run alone was consistent with this (chunk 1 audio coherent, $0.805 \rightarrow 0.985$, with no audio buffer present), but could not isolate the cause from the seeded audio noise, which is an equally plausible contributor and is present in every run.

The T2V run is the controlled test: identical seed and configuration, no guide image. Chunk 1 audio there is *also* coherent ($0.948 \rightarrow 0.969$, Table 5) — if anything more so than I2V’s. Holding the seed and audio-noise field fixed while removing the only visual anchor does not collapse chunk-1 coherence, so the hypothesis is not supported: audio coherence at chunk 1 tracks the seeded noise field and the audio generation process itself, not the presence of a visual prior. The two runs’ phase-lock behaviour argues the same direction from a different angle: I2V (with the guide image) is the more tightly locked run on *both* the audio-peak and the video-layout detectors (Section 7.6), which is the opposite of what a stabilising-anchor account would predict — if a strong visual prior constrained the joint generation, it should reduce repetition, not increase it. Subsequent chunks in both runs benefit from the audio SLB and hold within-chunk start coherence ≥ 0.93 (I2V) and mostly ≥ 0.88 (T2V) throughout.

7.4 Stability Margin

With $\rho_{\text{loop}} = 0.570$, the CLSS feedback loop provides a $\approx 43\%$ gain reduction per chunk before accounting for the model’s open-loop gain g . In the I2V run the spectral-shrinkage correction was mostly light — band-3 gains near 0.99 (a $\approx 3\%$ high-frequency energy excess over the EMA reference) for most chunks, rising to a $\approx 14\%$ excess at chunk 8 (gain 0.962). Taking the typical $\approx 3\%$ per-chunk growth as a rough estimate of the open-loop gain in the spectral channel gives $g \approx 1.03$ and $\rho_{\text{closed}} \approx 0.57 \times 1.03 \approx 0.59$, comfortably below the stability boundary of 1. The chunk-8 spike is the correction stack absorbing a transient (see Table 9) rather than a sign of runaway growth. The T2V run runs hotter on this channel: band-3 gains sit at 0.95–0.96 for most chunks, a $\approx 4\text{--}5\%$ typical excess, giving $g \approx 1.05$ and $\rho_{\text{closed}} \approx 0.57 \times 1.05 \approx 0.60$ — still comfortably stable, and close enough to the I2V estimate that the margin appears to be a property of the correction stack rather than of the conditioning mode. T2V’s raw audio, however, shows a spectral drift the correction stack does not reach: the Stage-1 trend summary’s **aud_hf** statistic (a ratio of high-band to reference energy, not covered by the video-side shrinkage of Section 3.3) grows from 1.26 to 1.89 over the run, with the top frequency bins individually reaching 2.1–2.65 $\times$ the chunk-1 reference by chunk 7 — I2V’s equivalent statistic is flat (0.87 $\rightarrow$ 0.87). This is audio-domain high-frequency energy growth with no analogue in the video channel or in the I2V run, and is unresolved by any current correction.

7.5 Stage-2 Identity Metric Ambiguity

Stage 2 lacks an anchor bank; its `identity_sim` reduces to comparing each window against the fixed first-window reference. On a single moving scene this conflates ordinary content motion with drift: in I2V, where the reference is the guide-image region, windows 2–7 sit at a 0.32–0.50 plateau, reflecting that the video has moved away from its opening frames, not instability. In T2V, where window 1 has no anchor to begin with, the same comparison instead oscillates near zero and slightly negative (–0.30 to +0.09 across windows 2–7) — consistent with window 1 itself being internally uncorrelated (Table 6, intra –0.231), so there is no stable reference for later windows to be “close to” or “far from” in the first place. Adding a scene-aware anchor bank to Stage 2 would make this metric interpretable in either conditioning mode; as reported it should be read only as “distance from the opening,” not as a drift signal.

7.6 Measured Failure Modes at Longer Horizons

Per-frame telemetry across extended runs — including the ten-chunk, ≈ 103 -second T2V and I2V runs reported above and shorter 13-lf runs — exposes two repetition mechanisms with distinct signatures, plus a Stage-2 audio seam behaviour.

Window-locked audio arcs. The audio energy peak locks to roughly the same relative window position across chunks: 94% of the window at 13-lf chunks (frame 102/109, eight consecutive chunks), and in the 31-lf I2V run the phase-lock detector

places the peak at relative frame 7 in most chunks (mode 7, covering 5/10 chunks within ± 1). The arc *scales with the window*, implicating the fed-forward conditioning geometry. Context envelope flattening (Section 5.3) was active in both runs and softens the lock without eliminating it: I2V’s chunk-to-chunk loudness-envelope correlation sits at 0.77–0.95 from chunk 3 onward, and its end-of-chunk within-chunk similarity declines across the run ($0.931 \rightarrow 0.785$), a second face of the same repetition. The T2V run — otherwise identical, differing only in the absence of a guide image — is the *less* locked of the two: its audio-peak mode covers only 3/10 chunks within ± 1 , and its envelope correlation swings between strongly negative (-0.49 to -0.27) and strongly positive (0.79 to 0.91) rather than settling into a lock. The guide image is therefore associated with *more* phase-lock, not less, which argues against reading the lock as an artifact of visual conditioning specifically; it is present, in weaker form, with no visual conditioning at all.

A ~ 4 -second video layout oscillation. Coarse scene layout departs from the scene reference and partially returns with a ≈ 4 s wall-clock period that is *invariant to chunk length*: at 13-lf (4.3 s) chunks the dip lands at new-frame 5–7 in every chunk (appearing seam-locked), while at 31-lf (9.9 s) chunks it recurs within the chunk at the same seconds-period. Context strength and context length (overlap 3/5/8) were tested and neither removes it; the oscillation appears intrinsic to the model’s trained temporal horizon. The overlap value trades the failure shape rather than curing it: in the overlap-8 I2V run the strict per-chunk swing-and-return is weaker (layout-argmin clustering loosens to mode 25, covering only 4/10 chunks within ± 1), but global layout still wanders and on some contents overlap 8 converts the oscillation into monotonic scene departure. The T2V run at the same overlap is looser still (mode 29, covering 2/10), and its largest layout events are spread across the whole run (chunks 1, 3, 6, 7, 9, 10) rather than concentrated at one seam as in I2V — the same guide-image/lock association seen in the audio signature above.

Stage-2 audio seams. With decoupled Stage-2 audio refinement, a high denoise fraction (σ -scale 1.0) re-rolls $\approx 45\%$ of the audio per window and the re-rolls disagree at every window boundary (boundary cosine 0.32–0.42) *regardless* of the audio SLB length — the SLB pins the seam frame, not the style of the re-roll. At denoise 0.4 the pass polishes instead: in the I2V run the refined audio stayed within seed similarity 0.79–0.81 of its Stage-1 source and the audio SLB was honoured at 1.000 at every window, yet seam similarity was still only 0.59–0.72 (worst at the first seam) — polishing narrows the disagreement but does not close it while windows are denoised independently. The T2V run, at the same denoise fraction and SLB honouring, seams noticeably better (0.75–0.84, also worst at the first seam): its Stage-1 audio boundary similarity was already higher (Table 5 vs Table 10), so Stage 2 starts each window’s polish from more agreeable source material. Rule: high-denoise audio rebuild only as a single window; multi-window refinement at ≤ 0.4 , and its seam quality inherits directly from Stage 1 audio boundary quality.

Unbounded audio high-frequency growth (T2V only). The T2V run’s top frequency bins grow from parity with the chunk-1 reference to $2.1\text{--}2.65\times$ it by chunk 7 (Section 6.2.3), a monotonic trend the trend-summary’s `aud_hf` statistic tracks rising from 1.26 to 1.89 over the run. The I2V run’s equivalent statistic is flat ($0.87 \rightarrow 0.87$). No current correction targets this: the video-side spectral shrinkage of Section 3.3 operates on the video latent only, and the audio-side corrections (Section 5.3) constrain global RMS and DC per channel but not the frequency distribution. Whether this is downstream of the higher overall audio loudness in T2V (Section 6.2.3) or a distinct effect of the missing visual conditioning is untested; an audio-side frequency-band shrinkage, mirroring Section 3.3 but applied to the audio latent, is the natural next mechanism to try.

8 Limitations

Single hyperparameter configuration, single seed. The reported T2V and I2V results share one set of hyperparameters (`new_lf` = 31, `overlap` = 8, 10 chunks, 1 scene), one seed, and one hardware configuration (16 GB VRAM, RTX-class); they differ only in the first-frame conditioning, by design, to isolate that variable. No baseline comparison against open-loop streaming is reported, and no ablation of the individual CLSS components (re-noising alone, AdaIN alone, shrinkage alone, anchor bank alone) or of the individual guidance terms (split CFG, modality, STG) has been performed. The paper therefore cannot quantify the marginal contribution of each correction or guidance term, or demonstrate improvement over the open-loop baseline from the same metrics, and every single-run comparison in Section 7 (e.g. the phase-lock/guide-image association) is a $n = 1$ -per-arm observation, not a statistically powered result. These comparisons are left to future work.

Stability metrics vs. perceptual quality. The reported metrics—boundary cosine similarity, intra-chunk cosine similarity, audio within-chunk segment similarity—measure the *stability* of the latent representation across chunks. They do not assess the perceptual quality of the generated content, prompt adherence, or realism. Standard quality metrics (FID, IS, CLIP similarity, human evaluation) would be required to characterise the generated video as a content generation system. High boundary similarity is a necessary but not sufficient condition for high video quality.

Object-level identity is unobserved. Every reported metric is computed from mean-pooled channel features or low-band spatial maps and is therefore *object-blind*: a vehicle or person can be replaced by a different instance across a chunk boundary while boundary and identity cosines remain above 0.95. This failure class was observed perceptually and is invisible to the telemetry. High-similarity readings must not be interpreted as object persistence. The available mitigation is textual: one prompt per chunk via `CLSSScenePrompts`, re-describing the persistent objects explicitly. Object-level tracking metrics (detection/re-identification on decoded frames) are future work.

Stability model assumptions. The linear gain model in Section 3.5 assumes that g is approximately constant and that the AdaIN and re-noising corrections act as independent multiplicative attenuators. Both assumptions simplify the analysis: the actual closed-loop dynamics involve higher-order interactions between corrections, guidance, and scene content. A more rigorous treatment (e.g., Lyapunov stability for the latent iteration) would require characterising the denoiser’s Jacobian across a distribution of latent inputs.

Hardware specificity. The GGUF 4-bit quantisation and CPU block-streaming configuration was designed for a specific VRAM budget (16 GB). Performance, latency, and numerical behaviour may differ on GPUs with more VRAM (where block-streaming is unnecessary) or with different quantisation schemes (e.g., Q8 or FP8).

9 Related Work

Long video generation. StreamingT2V [1] and FIFO-Diffusion [2] generate long videos by sliding-window denoising. Gen-L-Video [3] uses temporal attention patching. CLSS differs by applying closed-loop corrections specifically designed for the latent-streaming regime of large audio-video models.

Exposure bias in diffusion. The exposure bias problem in autoregressive models is well-studied [4]. For diffusion models, Ning et al. [5] and Salimans & Ho [6] address related distributional mismatch issues during training, but inference-time corrections for streaming have received less attention.

Statistical normalisation for consistency. Adaptive Instance Normalisation (AdaIN) [7] was introduced for style transfer, where the goal is to match the mean and standard deviation of a content feature map to those of a style reference. EMA-based statistical normalisation across video frames has also appeared in temporal consistency methods. In CLSS, AdaIN is repurposed as a *drift correction* applied to denoised latents between chunks, using the cross-chunk EMA as the style reference. The combination of EMA tracking with AdaIN blending (rather than full replacement) is motivated by the streaming setting, where the reference must itself evolve to allow intentional scene changes.

Frequency-domain video editing. Yang et al. [8] and Ceylan et al. [9] use frequency-domain constraints for temporal consistency in video editing. CLSS applies frequency shrinkage in the *latent* domain to a streaming generation setting, targeting energy accumulation rather than pixel-domain flicker.

10 Conclusion

We presented CLSS, a four-component closed-loop streaming synthesis framework that enables stable long audio-video generation with LTX-2.3 22B on consumer 16 GB VRAM hardware. The four corrections — calibrated context re-noising, EMA-tracked AdaIN, frequency-band soft shrinkage, and a dynamic anchor bank — achieve $\rho_{\text{loop}} = 0.570$; driven by a reference-parity multi-modal guider (split CFG, modality guidance, and STG) that correctly handles ComfyUI’s packed audio-video latent representation, the pipeline held Stage 1 video boundary similarity above 0.91 (I2V) and above 0.80 (T2V) across ten chunks of matched ≈ 103 -second single-scene clips, with the audio streaming buffer honoured at every seam in both and audio energy held to its anchored floor. A two-stage pipeline (streaming Stage 1 + chunked distilled Stage 2) delivers a fully refined $2\times$ -resolution audio-video output in approximately four hours per ≈ 100 -second clip on a single 16 GB GPU, essentially independent of conditioning mode.

Running matched T2V and I2V generations under an identical seed and guidance stack, differing only in the presence of a guide image, resolves a question the single I2V run could not: whether visual conditioning is responsible for chunk-1 audio coherence and for the tighter phase-lock observed in that run. It is not. T2V’s chunk-1 audio is coherent with no visual anchor at all ($0.948 \rightarrow 0.969$, matching I2V’s $0.805 \rightarrow 0.985$), and T2V is consistently the *less* phase-locked of the two runs on both the audio-peak and video-layout detectors — the opposite of what a stabilising-anchor account would predict. The guide image does, however, correlate with a substantially higher video boundary-similarity floor (0.910 vs. 0.797 at the same seam) and a lower audio RMS (0.42–0.50 vs. 0.59–0.78); T2V in turn shows an audio-domain failure mode absent from I2V — top-frequency energy growing to $2.1\text{--}2.65\times$ reference by chunk 7 — that no current correction targets. Two repetition mechanisms persist across both runs at this horizon and are characterised rather than resolved: a window-locked audio loudness gesture that envelope flattening softens but does not remove, and a chunk-length-invariant ≈ 4 -second video layout oscillation that appears intrinsic to the model’s trained temporal window. All of this is visible in the pipeline’s per-chunk telemetry — unconditional settings dumps, drift-event tables, and a phase-lock detector — which is designed to localise such effects in wall-clock time and latent-frame position, and to make a silently inactive guidance path impossible to miss (a guidance term that does not change per-step wall-clock time is not running).

Object-level identity persistence across chunk boundaries, invisible to the current structural metrics in both conditioning modes, remains the principal open problem for content-level quality.

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